Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/30/2026
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Residence Time Driven Flame Spread: The Final Phase		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	prof.bhattacharjee@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	619-594-6080
Organization Name:	San Diego State University		
PI Address 1:	5500 Campanile Drive, Mechanical Engineering Department		
PI Address 2:			
PI Web Page:			
City:	San Diego	**State:**	CA
Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	06/01/2021	**End Date:**	03/31/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2027 per F. Hernandez/NASA (Ed., 4/8/26). NOTE: End date changed to 03/31/2026 per D. Stocker/NASA-GRC (Ed., 9/30/25).		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), the list of Co-Investigators at San Diego State University has changed. Fletcher Miller, Ph.D. and Christopher Paolini, Ph.D. are no longer with the project. A new Co-Investigator is Luca Carmignani, Ph.D. He is based at San Diego State University (Ed., 4/1/26).		
COI Name (Institution):	Takahashi, Shuhei Ph.D. (Gifu University, Japan) Carmignani, Luca (San Diego State University)		
Grant/Contract No.:	80NSSC21K1126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of the RTDFS project is to investigate flame propagation over solid fuels in a microgravity environment. One of the central hypotheses is that steady flame spread over solid fuels cannot be sustained in a microgravity environment even at high oxygen environment if the fuel is thicker than a critical thickness. At lower oxygen level, a similar situation arises when the opposing flow velocity is below a critical velocity. A simplified theory based on scale analysis has produced closed-form formulas in terms of known parameters for these critical quantities. A comprehensive computational model that includes gas and surface radiation, finite-rate chemistry for pyrolysis and combustion support these results. Last year, a series of tests have been successfully conducted in the ISS. We are in the process of analyzing the massive amount of data collected from these tests performed over a span of six months, ending in February 2025. Preliminary analysis of the data seems to validate our hypothesis. We have submitted two manuscripts to the 41st International Symposium on Combustion, one of which has been accepted. A one year no-cost extension of the grant has been approved that will allow us sufficient time to fully analyze the data, prepare more manuscripts, and the final report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Preliminary analysis of the data collected from our recent ISS experiments confirms our hypothesis that flame spread cannot be sustained even for very thin fuels with a thickness greater than a critical thickness. This finding is expected to lead to a new definition of thermal thickness of solid combustibles in a microgravity environment. We have built three experimental setups at SDSU - the Flame Stabilizer, the Combustion Tunnel, and the Thin-Fuel Ignition Apparatus. These setups are being used by a new crop of students and a new faculty member in our department, producing valuable data on fuel ignition and flame propagation in normal gravity with implicaitons to fire safety on earth. The computational model for simulating ground based and microgravity flames is undergoing further refinemen with the goal of being released to the research community. Work is continuing to improve our thermodynamic portal, www.thermofluids.net, which is used by a huge number of students and educators around the world.
Task Progress:	Major Goals: The space-based experiments were intended to establish a critical fuel thickness above which steady flame spread cannot occur in a quiescent microgravity environment. The specific goals for this final phase of work are: (a) Download and catalogue the data from the NASA Glenn ISS Payload Operations Center (GIPOC); (b) Develop software to synthesize data from four different video cameras to reconstruct the flame spread process in each test; (c) Analyze the video data to extract the flame spread rate and other characteristics, such as flame size, pyrolysis length, etc. for each test; (d) Compare the experimental results with the theoretical predictions; (e) Investigate the pathway to extinction, especially the oscillatory behavior of the flame; (f) Continue with the development of theoretical and numerical models; (g) Continue the development of our educational outreach portal www.themofluids.net, currently used by students and educators around the world. Data Management: Downloaded over 1 TB of experiment files (images, sensor recordings, and telemetry data) from the GIPOC server to a secure Network Attached Storage (NAS) at the San Diego State University (SDSU) Fire Research Lab. Selected relevant images from the four cameras (SoFIE1, SoFIE2, SoFIE3, and ACME) for detailed analysis. Synchronized image timelines with sensor data by matching them with the ignition time (determined via ignition power-on signal and visual confirmation of the heating wire). Software Development: Developed MATLAB scripts to post-process still images (rotation, mirroring, and brightness/contrast adjustments) and generated composite videos to better visualize flame behavior and streamline image processing. Also, the Flame Tracker application (to analyze the flame images) has been updated and customized to handle the dim blue oscillating flames. Image Analysis: Processed experiment videos using Flame Tracker (Python-based software). Extracted key metrics (including leading-edge location, flame length, and flame area) using tracking methods based on RGB and HSV color spaces. Matched data from three adjacent cameras (SoFIE1, SoFIE2, and SoFIE3) using reference locations to provide a continuous view of flame behavior across the flat PMMA samples. Analyzed flame position over time for various PMMA thicknesses at 30% O2 across two pressure levels (0.5 and 1 atm). Observed that flames spreading over PMMA thicknesses exceeding 0.1 mm extinguished before reaching the sample ends. Theoretical Validation: Evaluated experimental results against previous theoretical predictions of extinction thickness in quiescent microgravity. Findings qualitatively align with the predicted limiting extinction thickness of 0.1 mm. Presented this work at the American Society for Gravitational and Space Research Conference (Phoenix, AZ, Dec 2025) and to be presented in the International Symposium on Combustion (Kyoto, Japan, July 2026).
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Kaplan S, Carmignani L, Bhattacharjee S, Ferkul P, Olson S, Stocker D. "Effect of fuel thickness and radiative losses on flame extinction in microgravity." 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. Abstracts. 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. , Dec-2025
Abstracts for Journals and Proceedings	Carmignani L, Bhattacharjee S, Kaplan S, Ferkul PV, Olson SL. "Near-limit flames in a quiescent microgravity environment." 2026 Spring Technical Meeting of the Western States Section of The Combustion Institute, Portland, Oregon, March 23-24, 2026. Abstracts. 2026 Spring Technical Meeting of the Western States Section of The Combustion Institute, Portland, Oregon, March 23-24, 2026. , Mar-2026

Papers from Meeting Proceedings	Carmignani L, Bhattacharjee S, Kaplan S, Ferkul PV, Olson SL. "Flame spread and extinction over acrylic sheets in microgravity at increased oxygen concentration." 41st International Symposium on Combustion, Kyoto, Japan, July 26-31, 2026. Abstracts. 41st International Symposium on Combustion, Kyoto, Japan, July 26-31, 2026. , Jul-2026
Papers from Meeting Proceedings	Bhattacharjee, S., Vera, E., Casebier, N. " An Ignition Apparatus to Characterize Ignition for Thermally Thin Fuels" 6th International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTC-2025), Jodhpur, India, December 9-12, 2025. Abstracts. 6th International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTC-2025), Jodhpur, India, December 9-12, 2025. , Dec-2025